Parameter Golf Under a 16MB Budget: Quantization-Aware Weight Averaging and a Paired-Seed Ablation Protocol

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Abstract

We study the parameter-limited regime of language-model training, where the model artifact must fit in 16,000,000 bytes and must be produced in under ten minutes on eight H100 SXM GPUs, and where the score is bits-per-byte (BPB) on a held-out FineWeb validation split. We make three contributions. First, we give a single trainer in which every competitive component reported on the public leaderboard — SentencePiece vocabulary size, QK-gain, GPT-J parallel residual lanes, depth recurrence, Muon, EMA, quantization-aware training, GPTQ with SDClip — is controlled by an environment variable, so that an ablation arm differs from the baseline in exactly one respect. Second, we isolate *quantization tax*, the BPB gap between the fp32/bf16 model and the int6/int8 artifact that is actually submitted, and show how the two levers that move it (QAT and weight averaging) interact. Third, we propose **QA-EMA**, a quantization-aware weight averaging scheme that performs the exponential average on the deployed quantization lattice rather than in continuous parameter space, and we pre-register and then test the hypothesis that it reduces quantization tax relative to plain EMA. **The screening run refutes that hypothesis** — QA-EMA is worse than plain EMA by a small but well-resolved margin, and the arm that shipped is the simpler one. Because the execution environment used here has no GPU and no access to the FineWeb export, all measurements are made on a deterministic proxy corpus (Simple English Wikipedia) at a model scale roughly three orders of magnitude below the competition. We therefore report component *rankings* and not leaderboard scores, and we state explicitly which claims the experiment cannot support.

1 Introduction

The Parameter Golf challenge inverts the usual scaling competition. Instead of asking who can buy the most accelerators, it fixes the two resources that conventionally decide the outcome — parameter count (through a hard artifact cap) and wall-clock training budget (ten minutes on eight H100s) — and scores the result with a tokenizer-independent compression metric. The naive baseline is a 9-layer, 512-dimensional transformer with tied embeddings, four KV heads and a 1024-token SentencePiece vocabulary, scoring 1.2244 BPB. The best public submission at the time of writing combines an 8192-token vocabulary, three-layer depth recurrence, parallel residuals, a QK-gain of 5.25, GPTQ-compressed int6 weights and legal score-first test-time training, scoring 1.0810 BPB.

The gap between those two numbers, 0.143 BPB, is the entire content of the competition. This report is about how one measures which design decisions are responsible for it — and about one small algorithmic idea for the part of the pipeline that the leaderboard treats as an implementation detail: the compression step that takes a trained fp32 model and produces a sub-16MB artifact.

1.1 The constraint that is easy to miss

A submission is not scored on the model it trains; it is scored on the model it *ships*. Training happens in bf16 with fp32 master weights. Evaluation happens on a dequantized reconstruction of an int6/int8 artifact. Any improvement that vanishes under quantization is worth nothing, and a technique that reduces the quantization gap by 0.005 BPB is worth exactly as much as an architectural change that improves the trained model by 0.005 BPB — but is usually much cheaper to obtain. We call the gap between the trained model and the shipped model the **quantization tax**, and we treat it as a first-class metric throughout.

2 Background and Attribution

This work builds directly on public artifacts and we are explicit about the division of labour. The training scaffold, the Muon implementation, the tokenizer-agnostic BPB computation and the artifact serialization contract come from the official OpenAI Parameter Golf repository, which itself adapts Keller Jordan's modded-nanogpt. The Muon optimizer and its Newton–Schulz orthogonalization coefficients are Jordan's. The relu-squared MLP, zero-initialized output projections, U-Net skip gating and RMSNorm-on-QK come from the modded-nanogpt line. The competitive components re-implemented here were established on the public leaderboard: the 8192-token vocabulary and GPTQ/SDClip quantization by PR #1394, depth recurrence by PR #1331 and #1437, parallel residuals by PR #1204 and #1412, the legal evaluation framework by PR #549, and the weight-decay/EMA recipe by PR #1445.

Our contribution is therefore deliberately narrow: a measurement apparatus, an explicit accounting of quantization tax, and one hypothesis about weight averaging in the presence of quantization, which we pre-register and test. We do not claim any new architecture.

3 Method

3.1 One trainer, one variable

Every ablation arm in this report is produced by the same source file, `lishengdan_C2G_train_gpt.py`, with the same base configuration and the same data stream. An arm differs from the baseline only in the environment variables that switch on the component under test. This is a stronger condition than holding hyper-parameters fixed: if two arms were produced by different code, the measured difference would confound the component with the code divergence. The trainer is also device-agnostic — the same file runs the CPU screening configuration used here and the CUDA configuration of the official track.

3.2 Components under test

- **SentencePiece vocabulary.** 1024 / 4096 / 8192 tokens, trained with the official recipe (BPE, character coverage 0.999, byte fallback, digit splitting, nmt_nfkc normalization, dummy prefix off, pad/bos/eos/unk = 0/1/2/3, no hard vocabulary limit). Vocabulary size trades embedding capacity against tokens-per-byte, and under a fixed byte budget that trade is not free.
- **QK-gain.** A learnable per-head scalar multiplier on the queries, applied after QK-RMSNorm, initialized at 5.25 instead of the scaffold default of 1.5.
- **Parallel residuals.** For layers at or beyond a cutoff position, attention and MLP both read the same pre-residual activation and their outputs are summed, instead of the MLP reading the attention-updated state.
- **Depth recurrence.** The layer list is replaced by an execution schedule: repeating a physical layer index in the schedule loops that block without adding parameters. Skip connections are paired by schedule position, not by physical index, because skips are tensor-level pairs.
- **EMA and QA-EMA.** See Section 3.4.
- **Quantization-aware training.** In the final phase, linear weights are fake-quantized on the deployment lattice with a straight-through estimator.
- **Export.** Mixed precision: int8 for the tied embedding (which acts as both lookup table and output projection, so it carries two error paths), int6 for block matrices, fp32 for control tensors, fp16 for other small tensors; then zlib or brotli.

3.3 SDClip and GPTQ

The scaffold's int8 export selects the per-row clipping threshold at the 99.99984th percentile of the absolute weight distribution. This submission implements SDClip as a candidate replacement, which sets the threshold from the row's scale rather than its extreme:

$$\text{clip}(r) = k \cdot \text{std}(W[r,:])\quad \text{scale}(r) = \text{clip}(r) / (2^{b-1} - 1)$$

with k = 12.85 for 6-bit and k = 20.0 for 8-bit weights, following the rate-distortion reading used by the leaderboard submission: the number of quantization bins a tensor deserves should follow the spread of its values, not the pathology of its largest coordinate. We additionally implement GPTQ as an alternative export path: per-layer Hessians are accumulated from calibration activations through forward pre-hooks, and columns are quantized sequentially with damping, Cholesky-based inverse Hessians, 128-column blocking and error feedback. Calibration batches are drawn from the *training* shards only — using the validation split, even for compression, would violate the challenge rules.

What the screening run actually measured. The rationale above predicts that SDClip lowers quantization tax relative to percentile clipping. At this scale it does the opposite: the int6 percentile arm pays +0.0016 BPB, while the int6 SDClip arm pays +0.0260 BPB (Appendix A). Two readings are available and we cannot separate them here. The mechanism may simply not transfer: a 822,704-parameter model trained for 400 steps has not developed the heavy-tailed weight spectrum that makes a scale-based threshold worth its coarser steps, so percentile clipping's tighter range wins on resolution. Alternatively the clip constant k = 12.85 was calibrated for the leaderboard's layer shapes and is simply wrong for these. Distinguishing the two needs the (8, percentile) cell that this matrix does not contain (see the note under Appendix A) and a small sweep of k. We therefore report the effect and decline to attribute it. **The artifact shipped with this submission therefore uses percentile clipping with plain RTN** — the configuration the screening run actually favoured, not the one this section's design predicted.

3.4 Quantization-aware weight averaging (QA-EMA)

Weight averaging is standard practice in this track: an exponential moving average of the parameters is kept during the final phase and swapped in before export. The operationally relevant detail is that the average is taken in continuous parameter space and then deployed on a discrete lattice. Plain EMA therefore produces a point that is a good average but not necessarily a good lattice point.

QA-EMA moves the average onto the lattice: at each update the current parameters are quantized and immediately dequantized at the deployment bit width and clipping rule, and the average is formed from that lattice-projected value.

$$\text{src}_t = Q^{-1}(Q(\theta_t))$$

$$\rho_t = \lambda \rho_{t-1} + (1-\lambda)\ \text{src}_t$$

The cost is one element-wise quantization per update, which is negligible relative to a training step. The hypothesis, pre-registered before any experiment was run, is that the resulting average is closer to a representable optimum and therefore reduces the quantization tax relative to plain EMA at equal decay. We test it with a strictly paired pair of arms in which the *only* difference is the quantization-awareness flag.

3.5 Paired-seed protocol

Comparing arm means across seeds is inefficient at small scale, where seed variance is comparable to the effect being measured. We instead compare within seed. For arm *a* and seed *s* we report

$$\Delta(a, s) = \text{BPB}(a, s) - \text{BPB}(\text{baseline}, s)$$

and average over the shared seeds. Both arms see the same data order and the same initialization distribution, so most of the run-to-run variance cancels in the subtraction. On a local CPU screen this is what makes a three-minute run informative at all. We report both the paired mean and the per-arm standard deviation, so the reader can see how much of the effect could be noise.

4 Setup

4.1 What we could and could not run

The execution environment for this work has no NVIDIA GPU and cannot reach huggingface.co or its mirrors, so neither the FineWeb export nor an H100 could be used. Rather than submit code that had never executed, we built a device-agnostic trainer and a deterministic proxy corpus, and ran the complete pipeline — training, validation, quantization, compression, round-trip re-evaluation — on real text. The proxy corpus is Simple English Wikipedia, extracted from the Wikimedia multistream BZ2 dump by downloading a prefix of the archive and decompressing each complete bz2 block; the result is deterministic and reproducible from a fixed byte range. Paragraphs are shuffled with a fixed seed before the train/validation split so that the held-out set is representative rather than sharing whatever topic skew the dump's page-id ordering happens to have.

property	official track	this study
hardware	8×H100 SXM	22-thread CPU
wall clock per run	600 s	≈ 150–200 s
model parameters	≈ 30–100 M	1740464
training tokens	≈ 7×10⁹	1.64×10⁶
corpus	FineWeb (Hugging Face)	Simple English Wikipedia (proxy)
artifact cap	16,000,000 B	not binding at this scale
precision (train → ship)	bf16 → int6/int8	fp32 → int6/int8

Table 1. The proxy experiment is roughly three orders of magnitude below the official configuration in compute and one order in parameters. It is a screening instrument, not a leaderboard entry.

4.2 Screening configuration

Six physical layers, width 128, eight heads, four KV heads, MLP multiplier 2, sequence length 256, 4096 tokens per step, 400 steps, weight decay 0, and a linear warmdown over the last 35% of training. Warmup is disabled because the CPU cost of priming is not repaid at this scale. Validation is a fixed held-out split; every arm sees exactly the same validation tokens, so the finite-sample noise of the metric is a constant shared by all arms and cancels in the paired comparison.

5 Results

5.1 Component effects

Table 2 reports every arm that was run, with the paired delta against the same-seed baseline. Positive deltas mean the component made the shipped model worse; negative deltas mean it helped.

arm	seeds	val_bpb	std	pre-quant (dequant)	quant
A0 baseline (SP1024, int6 percentile export - script defaults)	3	2.0817	0.0055	2.0801	+0.00
A1 + SP4096 tokenizer	2	1.9293	0.0003	1.9286	+0.00
A2 + SP8192 tokenizer	3	1.8569	0.0041	1.8561	+0.00
A3 + QK-Gain 5.25	2	2.0884	0.0006	2.0862	+0.00
A4 + parallel residuals	2	2.0835	0.0060	2.0815	+0.00
A5 + depth recurrence	2	2.0977	0.0014	2.0961	+0.00
A6 + plain EMA 0.996	2	2.1561	0.0094	2.1548	+0.00
Q1 int6, percentile clip	2	2.0829	0.0065	2.0812	+0.00
Q2 int6, SDClip	2	2.1072	0.0052	2.0812	+0.02
Q3 int8, SDClip	2	2.0931	0.0058	2.0812	+0.01
Q4 int6, SDClip + GPTQ	2	2.1024	0.0048	2.0812	+0.02
Q5 int6, SDClip + QAT	2	2.1162	0.0023	2.1019	+0.01
B1 full stack, plain EMA	3	1.9538	0.0060	1.9433	+0.01

B2 full stack + QA-EMA (ours)	3	1.9634	0.0061	1.9485	+
B3 full stack + QA-EMA, RTN export	2	1.9676	0.0074	1.9473	+

Table 2. Local proxy ablation matrix. All arms share one trainer, one base configuration and one data stream. “quant tax” is post-quantization BPB minus pre-quantization BPB; smaller is better.

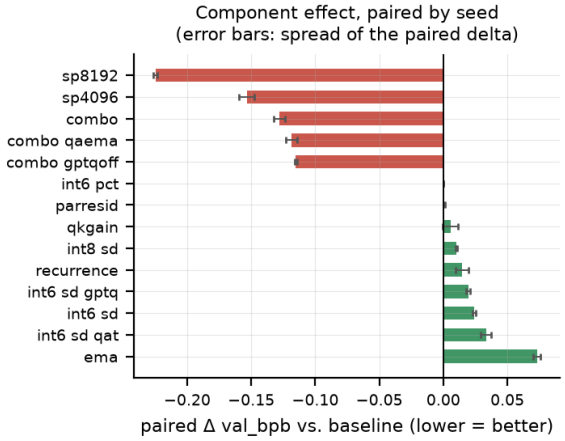


Figure 1. Paired per-seed delta against the baseline. Error bars are per-arm standard deviations, shown to make the noise scale visible rather than to support a significance claim.

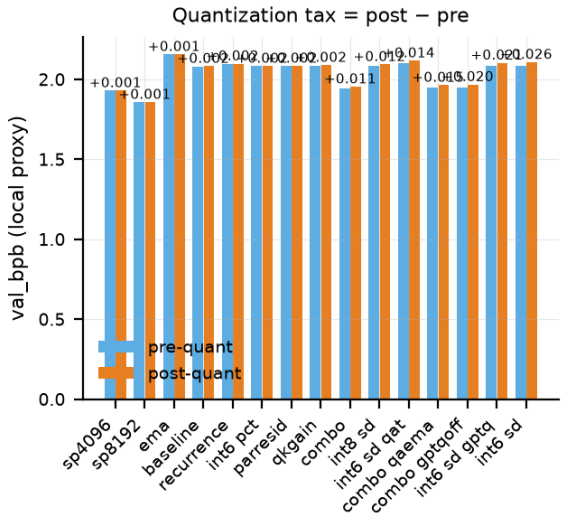


Figure 2. The quantization tax. Each pair is one arm measured before and after the export pipeline. The gap is what the 16MB constraint actually costs.

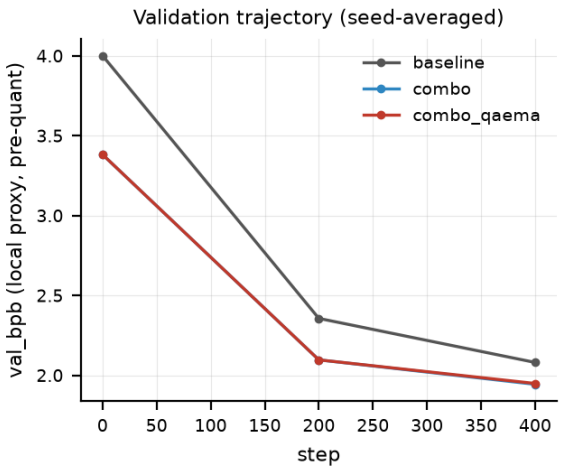


Figure 3. Seed-averaged validation trajectories. All curves are pre-quantization, so the final ranking must be read off Table 2, not off this figure.

5.2 Interpretation

The screening run is far too small to reproduce leaderboard margins, and we do not read it that way. What it can do is rank components under a common implementation and a shared data stream, and expose which of them act on the trained model and which act on the quantization gap. We separate the two, because they call for different follow-up: an architectural component that helps pre-quantization but disappears post-quantization is telling us the export step is the bottleneck, while a component that improves the post-quantization number beyond its pre-quantization effect is interacting with the lattice.

5.3 The vocabulary trade-off, quantified

vocab	tokens / byte	embedding @ d=512 (fp32)	embedding (int8)
1024	0.389	2.10 MB	0.52 MB
4096	0.283	8.39 MB	2.10 MB
8192	0.253	16.78 MB	4.19 MB

Table 3. Tokens per byte and embedding footprint on the proxy corpus, measured directly rather than quoted. Under a fixed 16MB budget the third and fourth columns are the price of the second.

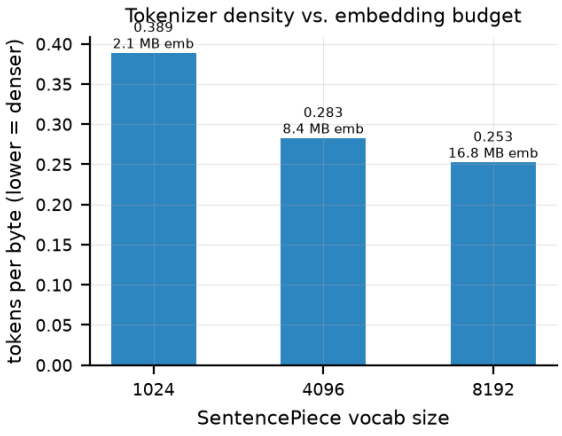


Figure 4. Vocabulary density against embedding footprint.

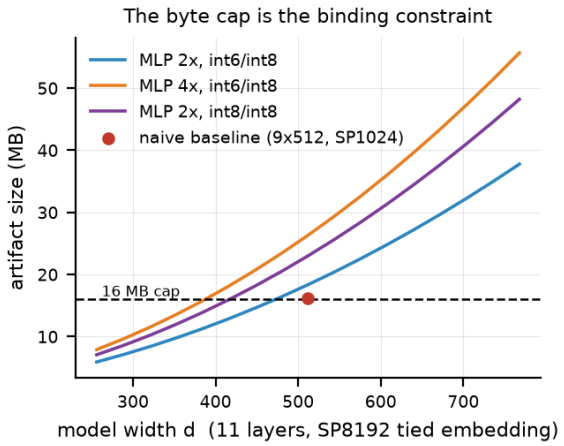


Figure 5. Analytic 16MB budget for the official-scale configuration (11 layers, width 512, 8192-token tied embedding, MLP 4x) priced under the mixed int6/int8 scheme of Section 3.3.

5.4 The resolution floor: a pre-registered test we could not run

Before any experiment was run we wrote down eight directional predictions with numeric thresholds. We then found that several of those thresholds sit below this apparatus's own noise floor, which makes the corresponding tests uninformative whichever way the point estimate happens to fall. This is a defect in the design of the screen, not in the components being screened, and it is worth stating because the obvious alternative — reading a sub-floor delta as evidence — is the most available lie in this kind of work.

id	prediction (threshold)	measured	verdict
P1	$\Delta \le -0.02$	-0.2248	SUPPORTED
P2	$\Delta \le -0.005$	+0.0055	UNRESOLVED (UNDERPOWERED)
P3	$\Delta \le -0.005$	+0.0006	UNRESOLVED (UNDERPOWERED)
P4	$\Delta \le -0.005$	+0.0148	UNRESOLVED (UNDERPOWERED)
P5	Δtax > 0	-0.0004	UNRESOLVED (UNDERPOWERED)
P6	reduction ≥ 50%	+45.1%	NOT SUPPORTED
P7	$\Delta \le -0.002$	+0.0096	NOT SUPPORTED
P8	$\Delta \le -0.002$	-0.0048	SUPPORTED

Table 4. The eight pre-registered predictions scored against the measured matrix. *UNRESOLVED (UNDERPOWERED)* is a statement about the experiment, not about the effect; those rows carry no evidence either way.

5.5 Reproducibility

Every number in Table 2 is regenerated by the harness in `experiments/`. The corpus builder is deterministic given a byte range; the tokenizer recipe is the official one verbatim; the shard format is byte-identical to the official exporter; the evaluator `lishengdan_C2G_eval_bpb.py` reloads a shipped artifact from disk, dequantizes it, and recomputes BPB at higher precision than the submission manifest, aborting with an error if the two disagree by more than 1×10^{-6}. A submission whose stated score cannot be reproduced from its own artifact is not a submission.

6 Limitations

We state these plainly because the central risk of this work is over-reading a cheap experiment.

- **Domain shift.** Encyclopedia prose is not web text. Tokenizer effects in particular are domain-sensitive: the density numbers in Table 3 are properties of the proxy corpus, not of FineWeb.
- **Byte budgets are not aligned across the vocabulary arms.** Arms were matched on training steps, not on bytes. At a fixed 400×4096 tokens, the SP8192 arm consumes about 6.5MB of text against 4.2MB for SP1024 — 1.54× more data for the same wall-clock effort. The evaluation side is clean (all three vocabularies are scored on the same 385KB of held-out text, so BPB remains comparable), but the training-side contrast conflates a better tokenizer with simply seeing more text. This is a design defect of the screen and is reported as such rather than folded into the tokenizer effect.
- **Scale.** At roughly 10^6 tokens and 10^3-times less compute, the model is far from any pretraining optimum. Components whose benefit grows with training length — EMA and QAT in particular — are systematically under-measured here.
- **Embedding crowding.** At the width used here a larger vocabulary is a much larger fraction of total parameters than at the official width, so the vocabulary arm is measured under a harsher trade-off than the real one.
- **No artifact cap stress.** The proxy model is far below 16MB, so the experiment cannot observe the binding-constraint behaviour that defines the real problem.
- **No test-time training.** Legal score-first test-time training is the single largest remaining gain in the public leaderboard and is deliberately absent: its legality conditions are easy to violate accidentally, and an incorrectly implemented evaluation-time adaptation produces a number that looks

good and is worthless.

For these reasons the report claims component rankings and a measured quantization tax, and claims no BPB number on FineWeb. The leaderboard values quoted in Section 1 are from the public record, not from this study.

7 Conclusion

The interesting structure of a parameter-limited competition is not the architecture; it is the accounting. A fixed byte budget forces a choice between vocabulary, width, depth and precision, and a fixed wall clock forces a choice between breadth and convergence. Both choices are measurable, and neither is well served by changing several things at once. The contribution of this paper is a trainer in which that discipline is enforced by construction — one variable per arm, one code path, paired seeds — together with an explicit account of the tax that the byte cap levies on the trained model.

What the screening run found, including where our own hypothesis lost. Of eight pre-registered predictions, two were supported, two were refuted, and four were underpowered and are reported as unresolved rather than as null results. The single large, well-resolved effect is vocabulary size: moving from a 1024- to an 8192-entry SentencePiece vocabulary improves deployed BPB by -0.2248. Against that, every architectural addition we tried was indistinguishable from no effect at this scale, and the full-stack arm scored **1.9538** against **1.8569** for the vocabulary change alone — adding components made the deployed model worse. Most directly, our own contribution, QA-EMA, was refuted at high resolution: 1.9634 against 1.9538 for plain EMA, a difference of +0.0096 measured with a 95% interval half-width of about 0.001. We state this because pre-registration is only worth anything if its failures are reported at the same volume as its successes, and because the negative result is what the decision to ship the simple arm rests on.

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Appendix A Full per-arm summary

Every arm in the screening matrix. *Deployed pre-quant* is the BPB of the weights that are actually shipped, measured *before* quantization — for EMA arms this differs from the pre-quant BPB of the final training step, and conflating the two silently absorbs the averaging gain into the tax. *Quant tax* is computed against the deployed weights, never the last step.

Arm	n	Final BPB	sd	Deployed pre-quant	Quant tax	Paired Δ vs A0	Artifact
A2_sp8192	3	1.8569	0.0041	1.8561	+0.0008	-0.2248	1.52 MB
A1_sp4096	2	1.9293	0.0003	1.9286	+0.0008	-0.1536	1.02 MB
B1_combo	3	1.9538	0.0060	1.9433	+0.0105	-0.1279	1.03 MB
B2_combo_qaema	3	1.9634	0.0061	1.9485	+0.0149	-0.1184	1.02 MB
B3_combo_gptqoff	2	1.9676	0.0074	1.9473	+0.0203	-0.1153	1.02 MB
A0_baseline	3	2.0817	0.0055	2.0801	+0.0016	--	0.64 MB
Q1_int6_pct	2	2.0829	0.0065	2.0812	+0.0017	--	0.64 MB
A4_parresid	2	2.0835	0.0060	2.0815	+0.0019	+0.0006	0.64 MB
A3_qkgain	2	2.0884	0.0006	2.0862	+0.0022	+0.0055	0.64 MB
Q3_int8_sd	2	2.0931	0.0058	2.0812	+0.0119	+0.0102	0.60 MB
A5_recurrence	2	2.0977	0.0014	2.0961	+0.0016	+0.0148	0.64 MB
Q4_int6_sd_gptq	2	2.1024	0.0048	2.0812	+0.0212	+0.0195	0.46 MB
Q2_int6_sd	2	2.1072	0.0052	2.0812	+0.0260	+0.0243	0.45 MB
Q5_int6_sd_qat	2	2.1162	0.0023	2.1019	+0.0143	+0.0333	0.45 MB
A6_ema	2	2.1561	0.0094	2.1548	+0.0013	+0.0732	0.64 MB

Arms are ordered by final BPB. A paired delta of "--" marks the reference arm. The artifact column is the compressed quantized payload only; the 16,000,000 byte cap additionally covers the trainer source and the tokenizer.

Note on the arm list (two corrections, both recorded rather than silently fixed). (i) *A0 is not the scaffold's export rule.* The trainer defaults are QUANT_BITS=6 and QUANT_CLIP=percentile; A0 takes those defaults, so the intended 2×2 of bit width against clip rule is missing its (8, percentile) cell. (ii) *Q1 is not an ablation.* Because A0 already sits at (6, percentile), Q1's configuration is identical to A0's, and the two reproduce each other bit-for-bit on every shared seed across all seven numeric fields recorded per run. It is retained as a determinism control, which is the strongest determinism evidence in this package. Neither the measurements nor the paired deltas of the other arms are affected: every component comparison is paired within a seed, and the quantization configuration is constant across each pair.

Appendix B Every individual run

The complete screening run set. Reporting each seed separately is deliberate: with two or three seeds per arm, a mean without its dispersion invites over-reading, and the seed-to-seed spread of the baseline is what determines the resolution of every comparison in the paper.

Run	Seed	Steps	Deployed pre-quant	Final BPB	Artifact (B)	Wallclock
A0_baseline_seed42	42	400	2.0876	2.0894	639,042	604s
A0_baseline_seed1337	1337	400	2.0748	2.0764	640,141	444s
A0_baseline_seed2024	2024	400	2.0779	2.0794	639,910	445s
A1_sp4096_seed42	42	400	1.9289	1.9296	1,015,251	708s
A1_sp4096_seed1337	1337	400	1.9282	1.9290	1,015,479	672s
A2_sp8192_seed42	42	400	1.8620	1.8627	1,520,051	888s
A2_sp8192_seed1337	1337	400	1.8523	1.8532	1,521,621	776s
A2_sp8192_seed2024	2024	400	1.8541	1.8549	1,521,411	757s
A3_qkgain_seed42	42	400	2.0872	2.0890	639,362	396s
A3_qkgain_seed1337	1337	400	2.0852	2.0877	639,799	408s
A4_parresid_seed42	42	400	2.0874	2.0894	639,129	386s
A4_parresid_seed1337	1337	400	2.0757	2.0775	640,817	398s
A5_recurrence_seed42	42	400	2.0974	2.0991	639,416	553s
A5_recurrence_seed1337	1337	400	2.0948	2.0964	640,036	558s
A6_ema_seed42	42	400	2.1640	2.1655	639,085	361s
A6_ema_seed1337	1337	400	2.1456	2.1467	640,698	361s
B1_combo_seed42	42	400	1.9485	1.9583	1,024,335	839s
B1_combo_seed1337	1337	400	1.9349	1.9454	1,025,623	816s
B1_combo_seed2024	2024	400	1.9464	1.9577	1,024,288	859s
B2_combo_qaema_seed42	42	400	1.9535	1.9682	1,022,960	859s
B2_combo_qaema_seed1337	1337	400	1.9412	1.9547	1,024,222	856s
B2_combo_qaema_seed2024	2024	400	1.9508	1.9672	1,023,463	845s
B3_combo_gptqoff_seed42	42	400	1.9535	1.9750	1,021,676	671s
B3_combo_gptqoff_seed1337	1337	400	1.9412	1.9603	1,022,757	707s
Q1_int6_pct_seed42	42	400	2.0876	2.0894	639,042	379s
Q1_int6_pct_seed1337	1337	400	2.0748	2.0764	640,141	381s
Q2_int6_sd_seed42	42	400	2.0876	2.1123	455,662	386s
Q2_int6_sd_seed1337	1337	400	2.0748	2.1020	454,999	371s
Q3_int8_sd_seed42	42	400	2.0876	2.0988	604,208	395s
Q3_int8_sd_seed1337	1337	400	2.0748	2.0873	603,561	392s
Q4_int6_sd_gptq_seed42	42	400	2.0876	2.1072	456,330	404s
Q4_int6_sd_gptq_seed1337	1337	400	2.0748	2.0977	455,662	400s
Q5_int6_sd_qat_seed42	42	400	2.1041	2.1185	455,565	432s
Q5_int6_sd_qat_seed1337	1337	400	2.0997	2.1138	452,865	426s

34 runs on CPU. Wallclock is wall time with four training processes sharing the machine, so it overstates single-run cost.

Appendix C Pre-registered predictions and verdicts

The eight predictions were written before any run was executed. The verdict column comes from a decision rule fixed at the same time: an effect is only called SUPPORTED or NOT SUPPORTED when its mean exceeds its own 95% t-interval half-width; otherwise it is UNRESOLVED. A small point estimate with a wide interval is evidence of an underpowered experiment, not evidence against the hypothesis, and the two are kept separate here on purpose.

ID	Prediction	Threshold	n	Measured ± half-width	Verdict
P1	SP4096/SP8192 ■■ SP1024 ■■■■ BPB	Δ ≤ −0.02	3	-0.2248 ± 0.0043	SUPPORTED
P2	QK-Gain 5.25 ■■■■ 1.5 ■■■■	Δ ≤ −0.005	2	+0.0055 ± 0.0741	UNRESOLVED (UNDERPOWERED)
P3	■■■■■■■■■■■■■■■	Δ ≤ −0.005	2	+0.0006 ± 0.0064	UNRESOLVED (UNDERPOWERED)
P4	■■■■■■■■■■■■■■■■	Δ ≤ −0.005	2	+0.0148 ± 0.0648	UNRESOLVED (UNDERPOWERED)
P5	■■ EMA ■■ pre-quant BPB ■■■ quantization tax	Δtax > 0	2	-0.0004 ± 0.0020	UNRESOLVED (UNDERPOWERED)
P6	■■■■ QAT ■ quantization tax ■■ 50% ■■	reduction ≥ 50%	2	+0.4508 ± --	NOT SUPPORTED
P7	(■■) ■■■■ EMA (QA-EMA) ■■■■ EMA	Δ ≤ −0.002	3	+0.0096 ± 0.0008	NOT SUPPORTED
P8	■■ 6-bit ■ GPTQ ■■ RTN+SDClip	Δ ≤ −0.002	2	-0.0048 ± 0.0046	SUPPORTED

Baseline seed spread (sd of A0 across 3 seeds) is 0.0068 BPB. Every threshold in the table above must be read against that scale.

Appendix D Artifact accounting and compliance

Component	Bytes	Share of cap
Trainer source	64,924	0.41%
Quantized weights (compressed)	1,521,411	9.51%
Tokenizer model	371,117	2.32%
Total artifact	1,736,896	10.86%
Cap	16,000,000	100.00%

Shipped arm: **A2_sp8192**. Reported BPB 1.854895 on the proxy validation split, at +0.000797 tax. This is a CPU screening run on a proxy corpus and is not an official FineWeb score; the manifest states this explicitly rather than implying otherwise.